



LA - Research Project

Organization

Oct 29, 2025

UNIVERSITÄT
**DUISBURG
ESSEN**

Offen im Denken

Allocations Overview

Topic	Group	Name
Topic 1 - Simulation Study using Student Models	Group 1	Muhammad Jalal Aden, Manjinder Singh, Mehmet Ali Yildirim
Topic 1 - Simulation Study using Student Models	Group 2	Komala Krishna Kasturi, Arpitkumar Ghelani, Karan Dineshbhai Ghoghari
Topic 2: Cognitive Load Analysis using Eye Tracking	Group 3	Sergei Lestsenski, Yonghao Chen, Alfred Fokoung Douo
Topic 3: Jupyter dashboard	Group 4	Arturo Olivares, Lisa Pawlowski
Topic 4: Automatic Concept Mapping and Knowledge Graph Generation	Group 5	Peisong Lin, Vincent Judyanto
Topic 4: Automatic Concept Mapping and Knowledge Graph Generation	Group 6	Hasti Lathiya, Shivangiben Lathiya, Abdelrahman Eldakrony
Topic 5: Semester Planning APP	Group 7	Priyank Bharatbhai Jivani, Angeline Patricia Kosasih, Axlkhan Suratman
Topic 5: Semester Planning APP	Group 8	Kevin Tchato Njami, Harianto Wahyu Wibowo, Pouya Zangooei
Topic 6: OATutor dashboard	Group 9	Reihaneh Sadat Hosseini Habibabadi, Parnian Afazeli, Mohammad Askari
Topic 6: OATutor dashboard	Group 10	Anjana Shaji, Sweetty Kheni, Waqar Ahmed
Topic 9: Revisiting the Grey Area approach of ZPD area with Bayesian Knowledge Tracing	Group 11	Sandeep Harikrishnan, Michael Breyer, Katjana-Antonia Kröhling

Note: Above research projects could be developed for potential publication in national or international conferences. We will provide support for you to present accepted papers there.

	Timeline			Check-In (Wed. 2pm.-4pm.): - Discuss the progress with your research advisors - Schedule a meeting time with your research advisors for checkin weekly.
	Week	Date	Milestone	
	1	15.10.25	Topic allocation	
	2	22.10.25	Topic allocation	
	3	29.10.25	Lab starts-intro & Literature review	In person
	4	05.11.25	Literature review & research proposal	In person
	5	12.11.25	Literature review & research proposal finalization	In person
	6	19.11.25	Methods design	In person
	7	26.11.25	Revisiting project proposals & Methods design	Presentation (Proposal)
	8	03.12.25	Implementation/analysis	In person
	9	10.12.25	Implementation/analysis	In person
	10	17.12.25	Implementation/analysis	In person
	11	24.12.25	Christmas holidays	
	12	31.12.25	Christmas holidays	
	13	07.01.26	Implementation/analysis	Presentation (progress check)
	14	14.01.26	Evaluation	In person
	15	21.01.26	Evaluation	In person
	16	28.01.26	Feedback & Documentation	In person
	17	04.02.26	Documentation	In person
	18	11.02.26	Documentation	In person
	19	18.02.26	Submission of final report	In person
	20	25.02.26	Exam	Final presentation

Research & Supervision

Topic	Research Advisor
Topic 1 - Simulation Study using Student Models	Mohamed Abdelmagied (mohamed.abdelmagied@stud.uni-due.de) Prof. Dr. Irene-Angelica Chounta (irene-angelica.chounta@uni-due.de)
Topic 2: Cognitive Load Analysis using Eye Tracking	Dr. Bibeg Limbu (bibeg.limbu@uni-due.de) Kaimao Sheng (kaimao.sheng@uni-due.de)
Topic 3: Jupyter dashboard	Mohamed Abdelmagied (mohamed.abdelmagied@stud.uni-due.de) Kaimao Sheng (kaimao.sheng@uni-due.de)
Topic 4: Automatic Concept Mapping and Knowledge Graph Generation	Mohamed Abdelmagied (mohamed.abdelmagied@stud.uni-due.de) Kaimao Sheng (kaimao.sheng@uni-due.de)
Topic 5: Semester Planning APP	Michael Dohmen (michael.dohmen@stud.uni-due.de) Prof. Dr. Irene-Angelica Chounta (irene-angelica.chounta@uni-due.de)
Topic 6: OATutor dashboard	Michael Dohmen (michael.dohmen@stud.uni-due.de) Kaimao Sheng (kaimao.sheng@uni-due.de)
Topic 9: Revisiting the Grey Area approach of ZPD area with Bayesian Knowledge Tracing	Michael Dohmen (michael.dohmen@stud.uni-due.de) Prof. Dr. Irene-Angelica Chounta (irene-angelica.chounta@uni-due.de)

Report/Paper submission

All research topics need to have the following chapter in the final report:

- Introduction
- Literature review
- Methods
- Results
- Discussion (if needed)
- Conclusion

up to 12000 words

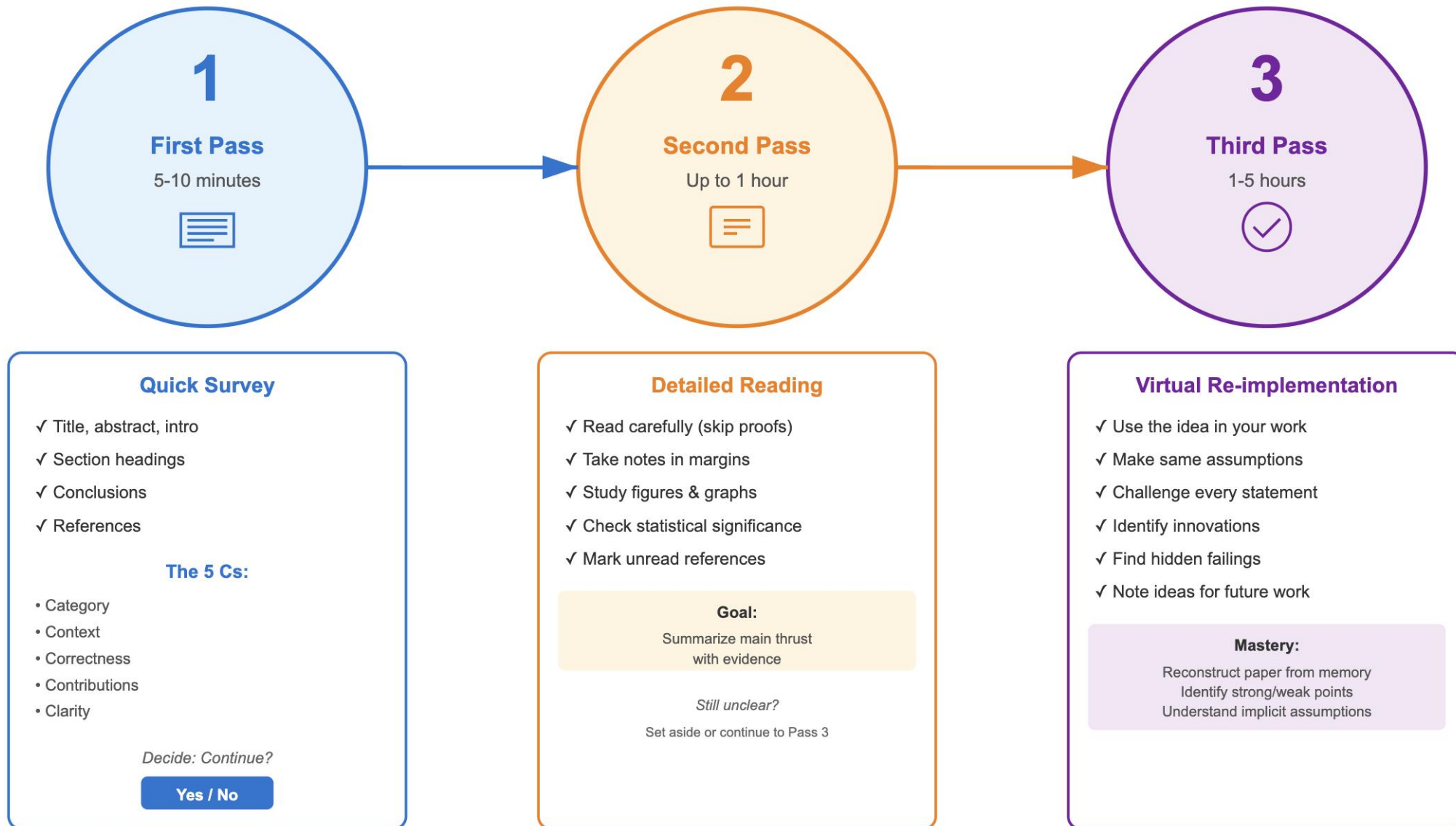
Please use the **LaTeX templates** for your writing. The templates have been uploaded to Moodle.

Final Presentation

All research topics need to have the following chapter in the final report:

- **Demo** (if necessary)
- **Introduction** (description of this project, background with limitations, research questions etc.)
- **Method** (technical details, experimental design etc.)
- **Results**
- **Conclusion**

Scientific Research Skills: How to read paper



How to read a paper:

(Srinivasan Keshav) <http://ccr.sigcomm.org/online/files/p83-keshavA.pdf>

Scientific Research Skills: How to write a paper

How to write a great research paper:

Simon Peyton Jones

<https://www.microsoft.com/en-us/research/academic-program/write-great-research-paper/>

Writing technical articles:

Henning Schulzrinne

<https://www.cs.columbia.edu/~hgs/etc/writing-style.html>

Scientific Research Skills: How to prepare a presentation

How to give a great presentation:

Simon Peyton Jones

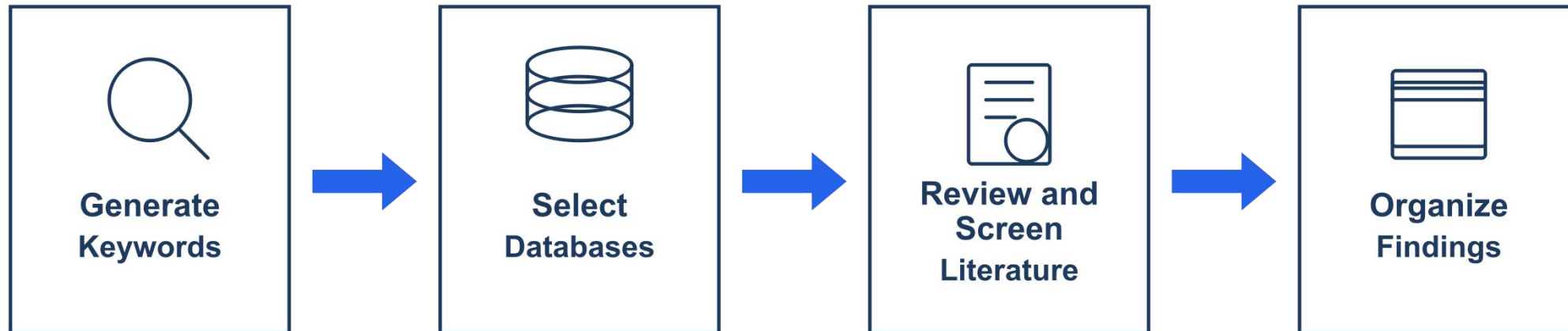
<https://www.microsoft.com/en-us/research/academic-program/give-great-research-talk/>

Scientific Research Skills: How to conduct a simple literature review

Database examples:

<https://dl.acm.org/>
<https://www.ebsco.com/>
<https://ieeexplore.ieee.org/>
<https://www.webofscience.com/>
<https://app.dimensions.ai/>

Check slide [How to read paper](#)



Check your research topic

Generate keywords

- Brainstorm synonyms, related terms, and acronyms.
- Use Boolean operators (AND, OR, NOT) to combine keywords.
 - Example: "machine learning" AND "education"

- Use reference management tools (such as Zotero, EndNote)
- **Group papers** by themes, methods, or outcomes

For your research proposal: What to do next?

Write a Research protocol:

The purpose of a research protocol is to provide a detailed plan that outlines exactly how a research study will be conducted. It serves as a blueprint for the entire research process — ensuring that the study is systematic, transparent, ethical, and reproducible.

Please check the research protocol template in the moodle.



Thank you!

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